DS002R - HW 3 - Wrangling

due Tuesday, September 22, 2026

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Introduction

In this assignment, we dive into using data wrangling to summarize datasets quantitatively and to format datasets so that they can be visualized.

Back to the Midwest... this time, wrangling.

In this part you will revisit and build on some of your findings from HW 2, where you explored the `midwest` data frame. Remember that the data frame is bundled with the **ggplot2** package and is automatically loaded when you load the tidyverse package. As a refresher, the data contains demographic characteristics of counties in the Midwest region of the United States. You can read documentation for the data set, including variable definitions by typing `?midwest` in the Console or searching for `midwest` in the Help pane.

> **Note**
>
> The data in the `midwest` data frame are from the 2000 census, so the information you compute below is likely slightly different from the values today.

Question 1

Calculate the number of counties in each state and display your results in descending order of number of counties. Which state has the highest number of counties, and how many? Which state has the lowest number, and how many?

> **Note**
>
> Commands Without Chaining by_state <- group_by(midwest, state) count_by_state <- summarize(by_state, count = n()) count_by_state_sorted <- arrange(count_by_state, desc(count)) view(count_by_state_sorted)

With Chaining

```
midwest |>
group_by(state) |>
summarize(count = n()) |>
arrange(desc(count))
```

```
# A tibble: 5 x 2
  state count
  <chr> <int>
1 IL      102
2 IN       92
3 OH       88
4 MI       83
5 WI       72
```

ANSWER: Illinois has the the highest number of counties (102) and Wisconsin has the lowest (72)

Question 2

Within a state, two counties can't have the same name, but across states county names can be shared. A friend says "Look, there is a county called XYZ in EVERY state in this dataset!" In a single pipeline, discover all counties that could fill in the value of XYZ. Your output should be a data frame with two columns: county and the number of times they appear in the data.

> Tip
>
> You will want to use the `filter()` function in your answer, which requires a logical condition to describe what you want to filter for. For example `filter(x > 2)` means filter for values of `x` greater than 2, `filter(y <= 3)` means filter for values of y less than or equal to 3.
>
> | operator | definition |
> |---|---|
> | `<` | less than |
> | `<=` | less than or equal to |
> | `>` | greater than |
> | `>=` | greater than or equal to |
> | `==` | exactly equal to |
> | `!=` | not equal to |

`x & y`	x AND y
`x | y`	x OR y
`is.na(x)`	test if x is NA
`!is.na(x)`	test if x is not NA
`x %in% y`	test if x is in y
`!(x %in% y)`	test if x is not in y
`!x`	not x

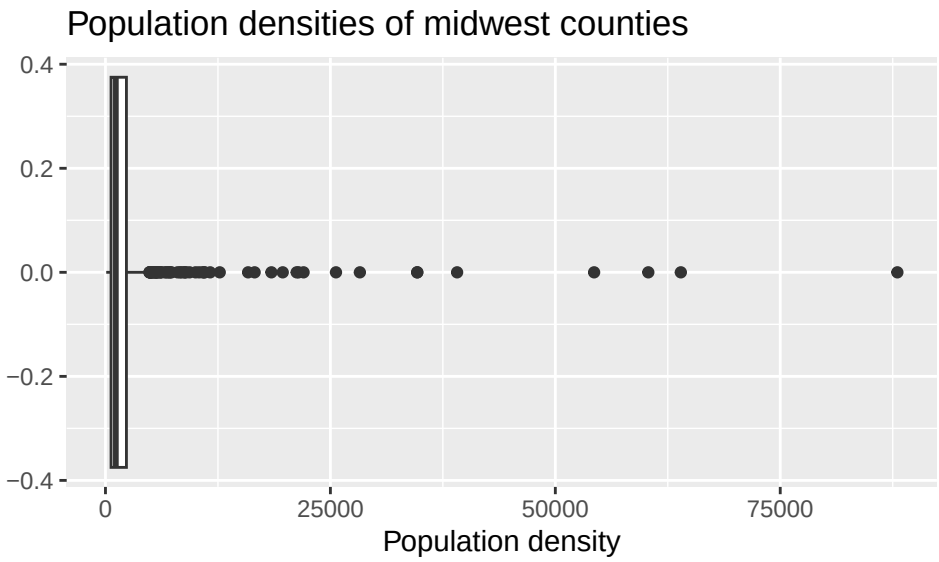
The table above is a summary of logical operators and how to articulate them in R.

```
midwest |>
 group_by(county) |>
 summarize(count = n()) |>
 filter(count >= 2) |>
 arrange(desc(count))
```

```
# A tibble: 70 x 2
   county     count
   <chr>      <int>
 1 CRAWFORD       5
 2 JACKSON        5
 3 MONROE         5
 4 ADAMS          4
 5 BROWN          4
 6 CLARK          4
 7 CLINTON        4
 8 JEFFERSON      4
 9 LAKE           4
10 WASHINGTON     4
# i 60 more rows
```

Question 3

Consider the following box plot of population densities where you can see that some counties have extreme values.



Identify the counties described in each part:

a. The counties with a population density higher than 25,000. Your code must use the `filter()` function.

```
midwest |>
filter(popdensity >= 25000) |>
select(county,state,popdensity,poptotal,area) |>
arrange(desc(popdensity))
```

```
# A tibble: 9 x 5
  county    state popdensity poptotal  area
  <chr>     <chr>      <dbl>    <int> <dbl>
1 COOK      IL        88018.  5105067 0.058
2 MILWAUKEE WI        63952.   959275 0.015
3 WAYNE     MI        60334.  2111687 0.035
4 CUYAHOGA  OH        54313.  1412140 0.026
5 DU PAGE   IL        39083.   781666 0.02
6 MARION    IN        34659.   797159 0.023
7 HAMILTON  OH        34649.   866228 0.025
8 FRANKLIN  OH        28278.   961437 0.034
9 MACOMB    MI        25621.   717400 0.028
```

b. The county with the highest population density. Your code must use the `max()` function.

```
midwest |>
filter (popdensity == max(popdensity)) |>
select(county,state,popdensity,poptotal,area)
```

```
# A tibble: 1 x 5
  county state popdensity poptotal  area
  <chr>  <chr>      <dbl>    <int> <dbl>
1 COOK   IL        88018.  5105067 0.058
```

> **Note**
>
> Answer using a **single** data wrangling pipeline for each part. Your response should be a data frame with five columns: county name, state name, population density, total population, and area, in order. If your response has multiple rows, the data frame should be arranged in descending order of population density.

Question 4

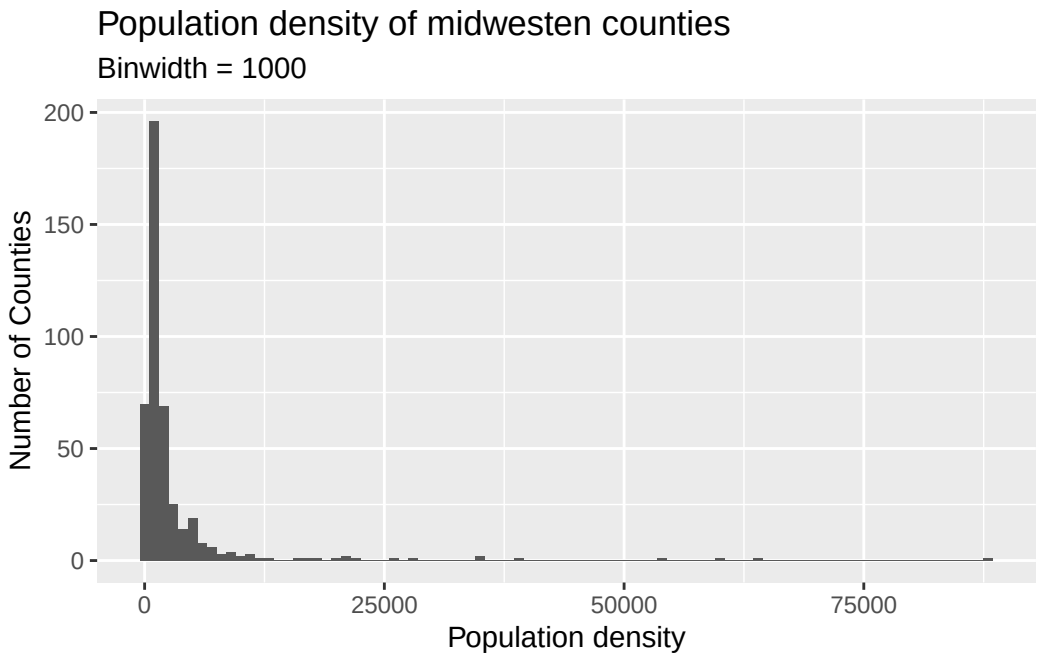
Let's say that you want to describe the distribution of population densities. The following is one acceptable description (in words) that touches on shape, center, and spread of this distribution. Create a plot that fits the description, and use data verbs to calculate the values that should go into the blanks. [You might want to look up the `quantile()` function by typing `?quantile` into the Console.]

> The distribution of population density of counties is unimodal and extremely right-skewed. A typical Midwestern county has population density of _____ people per unit area. The middle 50% of the counties have population densities between _____ to _____ people per unit area.

> **Tip**
>
> Think about which measures of center and spread are appropriate for skewed distributions. That is, there is a right answer for the choice of functions you use to calculate the center and spread.

```
ggplot(midwest, aes(x = popdensity)) +
geom_histogram(binwidth = 1000) +
labs(
x = "Population density",
y = "Number of Counties",
```

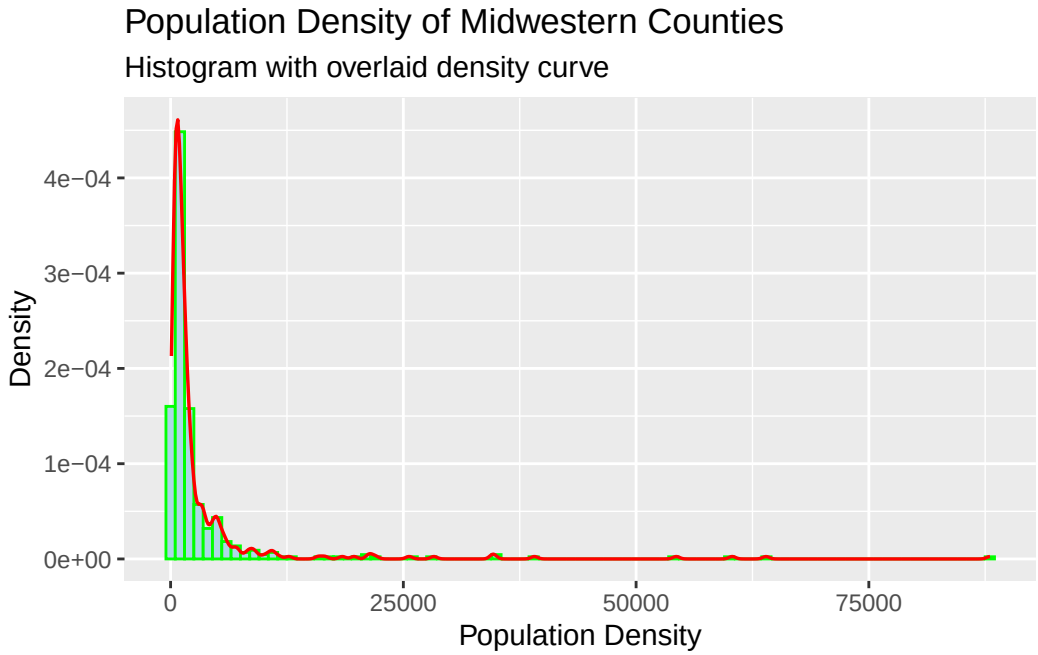
```
title = "Population density of midwesten counties",
subtitle = "Binwidth = 1000"
)
```



> **Note**
>
> Remember that a right-skewed or right-tailed distribution leans to the left and is said to have positive skew.

Another option no so pretty.

```
ggplot(midwest, aes(x = popdensity, y = after_stat(density))) +
  geom_histogram(binwidth = 1000, fill = "lightblue", color = "green") +
  geom_density(color = "red", linewidth = 0.6) +
  labs(
    x = "Population Density",
    y = "Density",
    title = "Population Density of Midwestern Counties",
    subtitle = "Histogram with overlaid density curve"
  )
```




```
midwest |>
  # select(popdensity) |>
  pull(popdensity) |>
  quantile(probs = seq(0,1,0.25), na.rm = FALSE)
```

```
      0%      25%      50%      75%     100%
  85.0500 622.4074 1156.2083 2330.0000 88018.3966
```

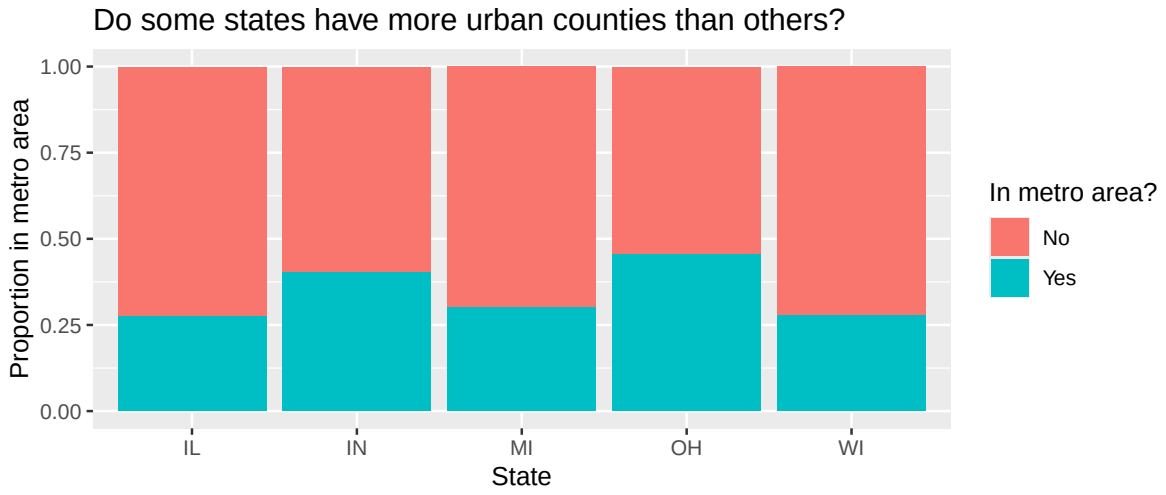
ANSWER: The distribution of population density of counties is unimodal and extremely right-skewed. A typical Midwestern county has population density of 1156 people per unit area. The middle 50% of the counties have population densities between 622 to 2330 people per unit area.

Question 5

Recall the visualization from HW 2 that showed the proportion of urban counties in each state.

```
midwest <- midwest |>
  mutate(metro = if_else(inmetro == 1, "Yes", "No"))
```

```
ggplot(midwest, aes(x = state, fill = metro)) +
  geom_bar(position = "fill") +
  labs(
    title = "Do some states have more urban counties than others?",
    x = "State",
    y = "Proportion in metro area",
    fill = "In metro area?"
  )
```



In a single data pipeline, calculate the proportion of counties in a metro area for each state (that is, the turquoise number from the plot above).

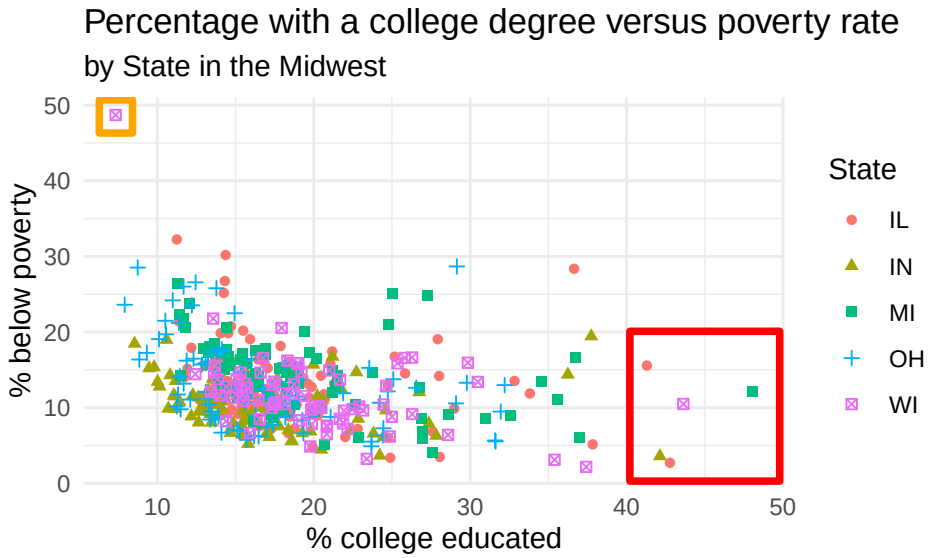
```
midwest |>
  mutate(metro = if_else(inmetro == 1, "Yes", "No")) |>
  group_by(state)  |>
  summarize(
    yes_count = sum(metro == "Yes"),
    no_count  = sum(metro == "No"),
    ratio  = yes_count / (yes_count + no_count)) |>
    arrange(desc(ratio))
```

```
# A tibble: 5 x 4
  state yes_count no_count ratio
  <chr>     <int>    <int> <dbl>
1 OH           40       48 0.455
2 IN           37       55 0.402
3 MI           25       58 0.301
```

```
4 WI            20        52 0.278
5 IL            28        74 0.275
```

Question 6

Return to the following scatter plot of percentage below poverty vs. percentage of people with a college degree, where the color and shape of points are determined by `state`. Recall that you were asked to identify (by name) at least one county that is a clear outlier.



a. In a single pipeline, identify the observations marked in the orange square on the upper left corner. Your answer should be a data frame with four variables: county, state, percentage below poverty, and percentage college educated.

```
midwest |>
  filter(percbelowpoverty >= 40) |>
  select(county,state,percbelowpoverty,percollege)
```

```
# A tibble: 1 x 4
  county    state percbelowpoverty percollege
  <chr>     <chr>            <dbl>      <dbl>
1 MENOMINEE WI                48.7       7.34
```

b. In a single pipeline, identify the observations marked in the red square in the plot above. Your answer should again be a data frame with four variables: county, state, percentage below poverty, and percentage college educated.

```
midwest |>
  filter(percollege >= 40) |>
  select(county,state,percbelowpoverty,percollege)
```

```
# A tibble: 5 x 4
  county    state percbelowpoverty percollege
  <chr>     <chr>            <dbl>      <dbl>
1 CHAMPAIGN IL               15.6        41.3
2 DU PAGE   IL                2.71       42.8
3 HAMILTON  IN                3.59       42.1
4 WASHTENAW MI               12.2        48.1
5 DANE      WI               10.5        43.6
```

c. Bring your answers from part (a) and part (b) together! In a single pipeline, and a single `filter()` statement, identify the observations marked in the red and orange square in the plot above. Your answer should again be a data frame with four variables: county, state, percentage below poverty, and percentage college educated.

```
midwest |>
  filter(percollege >= 40 | percbelowpoverty >= 40) |>
  select(county,state,percbelowpoverty,percollege)
```

```
# A tibble: 6 x 4
  county    state percbelowpoverty percollege
  <chr>     <chr>            <dbl>      <dbl>
1 CHAMPAIGN IL               15.6        41.3
2 DU PAGE   IL                2.71       42.8
3 HAMILTON  IN                3.59       42.1
4 WASHTENAW MI               12.2        48.1
5 DANE      WI               10.5        43.6
6 MENOMINEE WI               48.7         7.34
```

d. Create a new variable in `midwest` called `potential_outlier`. This variable should take on the value:

- `Yes` if the point is one the ones you identified in part (c), i.e., one of the points marked in the squares in the plot above.
- `No` otherwise.

Then, display the updated `midwest` data frame, with county, state, percentage below poverty, percentage college educated, `potential_outlier` as the selected variables, arranged in ascending order of `potential_outlier`. (Hint: look above and/or recall how `metro` was calculated using the function `if_else()`.)

```
midwest |>
  mutate(potential_outlier = if_else(percbelowpoverty >=40 | percollege >= 40, "Yes", "No")) |>
  arrange(desc(potential_outlier)) |>
  select(county,state,potential_outlier)
```

```
# A tibble: 437 x 3
   county    state potential_outlier
   <chr>     <chr> <chr>
 1 CHAMPAIGN IL    Yes
 2 DU PAGE   IL    Yes
 3 HAMILTON  IN    Yes
 4 WASHTENAW MI    Yes
 5 DANE      WI    Yes
 6 MENOMINEE WI    Yes
 7 ADAMS     IL    No
 8 ALEXANDER IL    No
 9 BOND      IL    No
10 BOONE     IL    No
# i 427 more rows
```

e. Recreate the visualization above, i.e., a scatterplot of percentage below poverty vs. percentage of people with a college degree, however color the points by the newly created `potential_outlier` variable instead of by state.

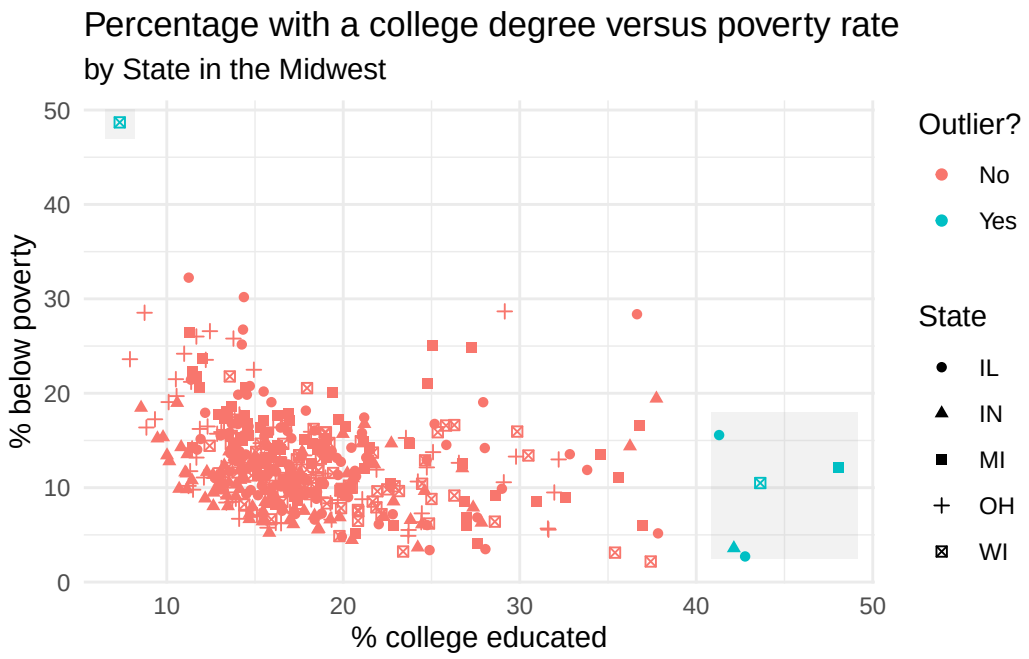
```
midwest2 <- midwest |>
  mutate(potential_outlier = if_else(percbelowpoverty >=40 | percollege >= 40, "Yes", "No")) |>
  arrange(desc(potential_outlier))
head(midwest2)
```

```
# A tibble: 6 x 30
    PID county   state  area poptotal popdensity popwhite popblack popamerindian
  <int> <chr>    <chr> <dbl>    <int>      <dbl>    <int>    <int>         <int>
1   570 CHAMPAI~ IL    0.058   173025      2983.   146506    16559           331
2   582 DU PAGE  IL    0.02    781666     39083.   714905    15462           962
3   691 HAMILTON IN    0.024   108936      4539    106764      676           163
4  1277 WASHTEN~ MI    0.041   282937      6901.   236390    31720          1076
5  2993 DANE     WI    0.073   367085      5029.   344617    10511          1201
```

```
6  3020 MENOMIN~ WI    0.021     3890       185.      416        0          3469
# i 21 more variables: popasian <int>, popother <int>, percwhite <dbl>,
#   percblack <dbl>, percamerindan <dbl>, percasian <dbl>, percother <dbl>,
#   popadults <int>, perchsd <dbl>, percollege <dbl>, percprof <dbl>,
#   poppovertyknown <int>, percpovertyknown <dbl>, percbelowpoverty <dbl>,
#   percchildbelowpovert <dbl>, percadultpoverty <dbl>,
#   percelderlypoverty <dbl>, inmetro <int>, category <chr>, metro <chr>,
#   potential_outlier <chr>
```

```r
ggplot(
  midwest2,
  aes(
    x = percollege,
    y = percbelowpoverty,
    color = potential_outlier,
    shape = state
  )
) +
    annotate(
    geom = "point",
    x = 45,
    y = 10.2,
    color = "grey",
    shape = "square",
    size = 25,
    stroke = 1,
    alpha = 0.2
  ) +
  annotate(
    geom = "point",
    x = 7.35,
    y = 48.5,
    color = "grey",
    shape = "square",
    size = 4.5,
    stroke = 1,
    alpha = 0.2
  ) +
  geom_point() +
  labs(
    title = "Percentage with a college degree versus poverty rate",
    subtitle = "by State in the Midwest",
```

```
    x = "% college educated",
    y = "% below poverty",
    color = "Outlier?",
    shape = "State"
  )  +
  theme_minimal()
```



Question 7

a. In a single pipeline (with multiple data verbs), calculate the total population for each state and save the resulting data frame as `state_population` and display it in descending order of total population.

```
midwest |>
  group_by(state) |>
  summarize(state_population = sum(poptotal)) |>
  arrange(desc(state_population))
```

```
# A tibble: 5 x 2
  state state_population
  <chr>            <int>
```

```
1 IL            11430602
2 OH            10847115
3 MI             9295297
4 IN             5544159
5 WI             4891769
```

b. Then, in a separate pipeline, calculate the proportion of the total population in each state (e.g., find the percent of people living in WI out of all the people in the midwest) and, once again, display the results in descending order of proportion of population.

```
midwest |>
 mutate(midwest_pop = sum(poptotal)) |>
 group_by(state) |>
 summarize(perc_mdw_pop = sum(poptotal*100/midwest_pop))|>
 arrange(desc(perc_mdw_pop))
```

```
# A tibble: 5 x 2
  state perc_mdw_pop
  <chr>        <dbl>
1 IL            27.2
2 OH            25.8
3 MI            22.1
4 IN            13.2
5 WI            11.6
```

> **Tip**
>
> In answering parts (a) and (b), you'll create two new variables, one for total population and other for proportion of total proportion. Make sure to give them "reasonable" names – short but evocative.

c. Which Midwestern state is most populous and what percent of the Midwest population lives there? Which is the least populous and what percent lives there?

ANSWER: The most populous is Illinois with 27.2% of the pop[ulation of midwestern states. Wisconsin is the least populated with 11.6%of the toal.

Question 8

Calculate the average percentage below poverty for each state and save the resulting data frame as `state_poverty` with the columns `state` and `mean_percbelowpoverty`.

Then, in a new pipeline, display the `state_poverty` data frame in ascending order of `mean_percbelowpoverty`. Which state has the lowest average percentage below poverty across its counties? Which state has the highest average percentage below poverty across its counties?

```
 state_poverty <- midwest |>
  group_by(state) |>
  summarize(mean_percbelowpoverty = mean(percbelowpoverty))
state_poverty
```

```
# A tibble: 5 x 2
  state mean_percbelowpoverty
  <chr>                 <dbl>
1 IL                     13.1
2 IN                     10.3
3 MI                     14.2
4 OH                     13.0
5 WI                     11.9
```

```
 state_poverty |>
  arrange(desc(-mean_percbelowpoverty))
```

```
# A tibble: 5 x 2
  state mean_percbelowpoverty
  <chr>                 <dbl>
1 IN                     10.3
2 WI                     11.9
3 OH                     13.0
4 IL                     13.1
5 MI                     14.2
```

ANSWER: The Midwest state with the lowest average percentage below poverty across its counties is Indiana (~10%) and the state with the highest average below poverty is Michigan (~14%).